JAINDU SAMARANAYAKE

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EDUCATION

Bachelor of Science Honors in Electronic and Telecommunication Engineering March 2024 - Present
University of Moratuwa, Sri Lanka CGPA: 3.65/4.00

G.C.E. Advanced Level Examination 2020 - 2023
Central College, Kuliyapitiya Z-Score: 2.1186
Results: 3 'A' Passes (Combined Mathematics, Physics, Chemistry)

TECHNICAL SKILLS & INTERESTS

Hardware & Embedded	ESP32, Raspberry Pi, Arduino Mega, FPGA (DE0-Nano, DE2-115), I2C, UART
Engineering Software	Altium Designer, SolidWorks, OMNeT++, Quartus, MATLAB/Simulink
, Programming	C++, Python, Java
Domains of Interest	Embedded Systems, Firmware Development, IoT, Network Architecture, ML

HONOURS & AWARDS

- **4th Place** – SPARK Challenge: Recognized for creating sustainable, market-ready IoT solutions with the Dengue Patient Monitoring System (Team NEXORA).
- **Finalist** – Sri Lanka IoT Challenge (SLIoT): Developed the Dengue Patient Monitoring System (Team NEXORA).
- **Finalist** – Sri Lanka Robotics Challenge (SLRC) 2026.

CERTIFICATIONS

- **Deep Learning Onramp** – MathWorks Credential ↗
- **Machine Learning Onramp** – MathWorks Credential ↗
- **MATLAB Onramp** – MathWorks Credential ↗

PROJECTS

Dengue Patient Monitoring System (Team NEXORA) | *SLIoT & SPARK* November 2025 - Present

- Engineered a wrist-worn ESP32 telemetry node to continuously measure and transmit critical patient vitals (SpO2, blood pressure).
- Programmed I2C sensor protocols and established a 2-way serial communication link with a Raspberry Pi for data aggregation.
- Deployed a lightweight ML model directly onto hardware for sub-second latency analysis of patient vitals.

Smart Water Management System | *Academic Group Project*

- Designed a custom PCB buck-boost converter and LDO circuit using Altium Designer for stable 5V/3.3V logic distribution.
- Developed a digital water flow tracking device integrating a pulse meter with Hall effect sensors to monitor live consumption.
- Interfaced a GSM module to wirelessly transmit hardware telemetry to a cloud server with technical specifications.

Autonomous Robot Design (SLRC 2026) | *Competition Project* July 2025 - April 2026

- Architected a computer vision pipeline capable of decoding visual payloads into precise grid coordinates.
- Compiled a custom C++ library to process a 56,000+ entry mapping table, preventing 4GB RAM memory overflow crashes.
- Engineered a REST API bridge to sync real-time hardware telemetry and LED state indicators, achieving 100% uptime.

Laboratory Database Management System | *Academic Group Project*

- Developed an administrative frontend dashboard interfacing with a REST API to manage over 100 laboratory assets.
- Architected the backend using Spring Boot (Entity, Service, and Controller layers) with Spring Data JPA for database interactions.
- Implemented secure role-based access control (RBAC) using JWT authentication and managed project builds using Maven.

Hardware Robot Design (Line & Wall Following) | *Academic Project* Finished December 2025

- Designed and constructed a custom ball-shooting mechanism utilizing a rack and pinion system integrated with dual rotating wheels.
- Contributed to overall mechanical part design using SolidWorks, executing comprehensive hardware testing and system integration.

FPGA Co-Processor Accelerator | *Ongoing Academic Group Project* Present

- Researching hardware architectures to design a custom FPGA-based co-processor for acceleration using a DE2-115 board.
- Evaluating existing IP cores and libraries for optimized computer arithmetic operations to integrate into the accelerator.

FPGA UART Communication Protocol | *Academic Project*

- Implemented a full-duplex UART serial communication protocol at 115200 baud utilizing a DE0-Nano FPGA board.
- Configured data frame structure utilizing an 8-bit word data packet encapsulated with a start bit and stop bit.

Digital Hardware Product Development | *Academic Project* February 2025 - June 2025

- Executed full hardware lifecycle for an audio prototype, utilizing Altium Designer for a 2-layer PCB and SolidWorks for enclosure.

POSITIONS OF RESPONSIBILITY & EXTRACURRICULAR

- **Committee Member, IEEE Robotics and Automation Society (RAS)** November 2025 - Present
  Co-organized MAZEX 1.0 Micromouse competition; drafted outreach communications and coordinated mentor sessions.
- **Organizing Committee Member:** ENTC Career Fair '26 (Jan 2026) & SLRC (Apr 2025)
- **Mentor, Soyuru Sathkara Seminar Series** January 2025
  Conducted targeted mathematics seminars for G.C.E. Ordinary Level (O/L) students for examination preparation.

REFERENCES

Dr. Ranga Rodrigo Email: ranga@uom.lk
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